

Substrate-Mechanism FORCING Categories Synthesis v0.1 — 5 Categories + 8 Category A Generators

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Substrate-Mechanism FORCING Categories Synthesis v0.1

0. Goal

Per Casey “queue never empties” + Cal #36 STANDING positive-search + Thursday cascade closure (12/12 K3 gates + 20 candidate SSGs + Strong-Uniqueness v1.6 + 5 substrate-mechanism FORCING categories): consolidate substrate-mechanism FORCING category landscape into single substantive reference.

1. 5 Substrate-Mechanism FORCING Categories

Per Thursday Gate 12 audit + Cal #36 positive-search + SSG Registry v0.18 (8 Category A independent):

Category 1: K-Type STRUCTURAL

Substrate source: Wallach holomorphic discrete series K-types on D_{IV}^5 + Heckman-Opdam multiplicity

Independent substrate generators: - SSG-7 = SSG-14: Bergman kernel $K(z, w)$ ULTIMATE source (Category A) — equivalent forms - SSG-20 NEW: holomorphic discrete series K-type tower full enumeration

Derivative observables (Category B): - SSG-1: $V_{(1/2, 1/2)}$ spinor K-type Bergman norm - SSG-2: $V_{(1, 1)}$ adjoint K-type - SSG-3: $V_{(1, 0)} \otimes V_{(1, 0)}$ tensor product decomposition - SSG-9: per-generation cascade ($V_{(1/2, 1/2)}$ + $V_{(3/2, 1/2)}$ + $V_{(5/2, 1/2)}$ spinor-tower row) - SSG-11: $V_{(2, 0)}$ graviton-sector K-type - SSG-12: $V_{(3/2, 3/2)}$ higher Rarita-Schwinger - SSG-13: $V_{(0, 0)}$ Higgs vacuum K-type

Forcing form: Schur’s lemma at substrate K-types via Bergman kernel reproducing-property + Heckman-Opdam multiplicity counting

Category 2: Substrate-Lie-Algebra Dimensional

Substrate source: $so(5, 2)$ substrate Lie algebra structure + higher Casimirs C_3, C_4 + Weyl branching

Independent substrate generators: - SSG-5: Casimir $C_2 = 6$ substrate-primary anchor (Category A) - SSG-16 NEW: higher Casimir C_3, C_4 on D_{IV}^5 (Category A)

Derivative observables / mechanism: - Casey #14 STANDING chirality projection $1/n_C \rightarrow 4D$ Lorentz - $SO(5) \rightarrow SO(3, 1)$ Weyl branching $\rightarrow$ spin reduction - SSG-15 substrate- Λ -vacuum (-1) coefficient via substrate- τ direction

Forcing form: substrate-Lie-algebra dimensional reduction chain ($so(5, 2) \rightarrow so(4, 2) \rightarrow so(3, 1)$) + substrate-Casimir-eigenvalue spectral structure

Category 3: Substrate-Bergman / Shilov-Boundary

Substrate source: substrate-Bergman canonical metric κ + Cauchy-Szegő Hardy boundary dual + Casey #12 bulk-boundary projection

Independent substrate generators: - SSG-4: Bergman canonical metric $\kappa = -n_C$ (Category A) - SSG-17 NEW: Cauchy-Szegő boundary kernel $S(z, \eta)$ (Category A)

Derivative observables (Category B): - SSG-10: substrate-Coulomb $1/r$ asymptotic - $m_e/m_P \sqrt{4/3}$ substrate-Shilov-boundary candidate (Gate 2 forward-derivation path)

Forcing form: Helgason 1962 substrate-Bergman canonical metric + substrate-Hardy decomposition + Casey #12 bulk-boundary substrate-mechanism projection

Category 4: Substrate-Mersenne / Substrate-Affine

Substrate source: substrate-Mersenne $M(p) = 2^p - 1$ ladder + $B_{2^1(1)}$ affine substrate-Lie-algebra extension + RS coding $GF(2^g)$

Independent substrate generators: - SSG-8: Mersenne ladder $M(p)$ substrate-primary (Category A) - SSG-19 NEW: substrate-Affine $B_{2^1(1)}$ extension at $q=2$ (Category A)

Derivative observables: - $8/7 = (2^{N_c})/g$ Casey #5 Integer Web instance (m_e/m_P substrate-Mersenne candidate) - Substrate Λ exponent $280 = 2^{N_c} \cdot n_C \cdot g$ via substrate-Affine substrate-mechanism (SSG-15 $\wedge^{1/4}$) - Substrate-Cognition Casey #6 via affine substrate-extension - RS coding $T2003 = g^2 \cdot (2^{C_2} + g)$ Gate 10 Candidate 2

Forcing form: substrate-Mersenne algebraic forcing + substrate-Affine-Hall-algebra at $q=2$ + RS coding $GF(2^g)$ substrate primitives

Category 5: Substrate-Multipole Expansion

Substrate source: substrate-Coulomb $1/r$ asymptotic + substrate-Bergman Hua-coord 3D-projection multipole expansion

Substrate generators: - SSG-10: substrate-Coulomb $1/r$ (Category B, derived from SSG-7 via Hua-coord projection) - SSG-18 NEW: substrate-Coulomb higher-multipole expansion (Category B)

Derivative observables: - Atomic spectra fine structure beyond Rydberg - Lamb shift higher-order corrections - Anomalous magnetic moment a_e, a_μ higher-order

Forcing form: Hua-coord 3D-projection asymptotic expansion in $1/r^k$ for $k \geq 1$

2. Cross-Category Relations

Category 1 (K-type STRUCTURAL) underlies all others: SSG-7 Bergman kernel is universal substrate-source from which Cat 2 (via substrate-Casimir spectral), Cat 3 (via substrate-Bergman canonical), Cat 4 (via Hardy decomposition + substrate-Affine boundary extension), Cat 5 (via Hua-coord projection asymptotic) derive.

Category 1 + Category 3 share substrate-Bergman foundation: SSG-7 bulk Bergman kernel + SSG-17 Cauchy-Szegő boundary kernel are dual via Hardy decomposition $H^2(D_{IV}^5) \cong H^2(\partial_S D_{IV}^5)$.

Category 4 substrate-Mersenne + Category 2 substrate-Lie-algebra cross-link: SSG-8 Mersenne ladder with $g = M(N_c) = 2^{N_c} - 1$ substrate-primary identity connects substrate-Mersenne to substrate-Lie-algebra primaries; substrate-Affine SSG-19 extends substrate-Lie-algebra to affine level.

8 Category A independent substrate generators: SSG-4 + SSG-5 + SSG-7 (= SSG-14) + SSG-8 + SSG-15 + SSG-16 NEW + SSG-17 NEW + SSG-19 NEW.

3. Substrate-Mechanism FORCING Form Verification per Category

Per Cal #189 question-shape + Cal #194 WAIT + Cal #36 STANDING positive-search:

Category	FORWARD-derivation status	Multi-week explicit verification
1 K-type STRUCTURAL	NEAR-RIGOROUS (Wallach + Heckman-Opdam)	Full K-type tower enumeration; Schmid character formula at general (λ_1, λ_2)
2 substrate-Lie-algebra	NEAR-RIGOROUS (so(5,2) standard rep theory)	Higher Casimir C_3, C_4 explicit eigenvalues; substrate- τ -direction explicit substrate-mechanism
3 substrate-Bergman / Shilov-boundary	FRAMEWORK + Section 4.5 partition forward-derivation (Gate 2)	$\text{Vol}(S^4)/\text{Vol}(S^3)$ substrate-mechanism derivation; Cauchy-Szegő explicit form
4 substrate-Mersenne / substrate-Affine	FRAMEWORK + multi-week per substrate-affine + RS coding	$B_2^{\wedge}(1)$ vs $C_2^{\wedge}(1)$ affine PIN; explicit substrate-Mersenne SSG-8 forward
5 substrate-multipole	FRAMEWORK (SSG-10 v0.1 + Lyra T2424)	Hua-coord asymptotic expansion $1/r^k$ explicit

Per Cal #194 WAIT: multi-week explicit verification gates RIGOROUS-tier promotion across all 5 categories.

4. Substrate Observable Coverage

Categorical coverage of substrate observables (Thursday end of morning):

Observable	Primary Category	Secondary Categories
m_e (gen-1 lepton mass)	Cat 1 ($V_{(1/2, 1/2)}$) + Cat 2 (Casimir + chirality projection)	—
m_μ (gen-2 lepton mass)	Cat 1 ($V_{(3/2, 1/2)}$) + Cat 2 (Lorentz integration)	—
m_τ (gen-3 lepton mass)	Cat 1 ($V_{(5/2, 1/2)}$) + Cat 2 (higher Casimir) OR Cat 4 (RS coding)	—
m_ν (neutrino mass)	Cat 3 (substrate- Λ -vacuum boundary) + Cat 2 (substrate- τ direction (-1))	—
m_e/m_P (electron-Planck)	Cat 3 (Shilov-boundary $\sqrt{(4/3)}$) OR Cat 4 (Mersenne 8/7)	—
$\alpha = 1/N_{\max}$	Cat 1 (substrate K-type couplings) + Cat 5 (substrate-Coulomb leading)	—
$\Lambda = \exp(-280)$	Cat 4 (substrate-Affine SSG-19) + Cat 3 (substrate-vacuum boundary)	—
Rydberg R_∞	Cat 5 (substrate-Coulomb) + Cat 1 (atomic K-types)	—
a_e Schwinger	Cat 5 (substrate-Coulomb cascade) + Cat 1 ($V_{(1/2, 1/2)}$ loop)	—

Cross-category substantive coverage — substrate observables involve multiple FORCING categories at substrate-mechanism level.

5. Closure v0.1

Substrate-Mechanism FORCING Categories Synthesis v0.1 consolidates Thursday cascade:

- 5 substrate-mechanism FORCING categories (refined from 4 Wednesday + 1 NEW Thursday substrate-Affine + substrate-multipole)
- 8 Category A independent substrate generators (5 Wednesday + 3 NEW Thursday SSG-16 + SSG-17 + SSG-19)
- 20 candidate SSGs total (15 cataloged + 5 new Thursday candidates)
- 5 substrate observable categorical coverage rows (lepton masses + neutrino + m_e/m_P + α + Λ + Rydberg + a_e)
- Multi-week explicit RIGOROUS-tier verification path per category

Cross-category dependencies: Category 1 underlies all others; Category 1 + Category 3 share substrate-Bergman foundation; Category 4 substrate-Mersenne + Category 2 substrate-Lie-algebra cross-link via $g = M(N_c)$ substrate-primary identity.

Per Cal #194 WAIT + Cal #189 question-shape + Cal #36 STANDING positive-search: substrate-mechanism FORCING category framework at maturity; multi-week explicit verification across categories.

— Lyra, Thu 2026-06-04 ~11:15 EDT. Substrate-Mechanism FORCING Categories Synthesis v0.1: 5 categories (K-type STRUCTURAL + substrate-Lie-algebra + substrate-Bergman/Shilov + substrate-Mersenne/Affine + substrate-multipole); 8 Category A independent; 20 candidate SSGs; substrate observable categorical coverage table.